

WATERBURY-OXFORD, CT, USA

WMO: 725029

Lat: 41.483N

Lon: 73.133W

Elev: 221

StdP: 98.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-13.0	-24.0	0.4	-13.6	-21.7	0.6	-11.6	11.1	-3.2	9.3	-4.0	3.3	330	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.9	23.1	28.9	22.0	27.5	21.1	24.4	28.7	23.4	27.1	22.6	25.8	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	26.3	22.4	17.5	25.6	21.6	16.7	24.6	74.9	28.5	70.7	27.0	67.6	25.7	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	6.8	-19.6	33.0	2.5	1.6	-21.4	34.1	-22.9	35.1	-24.3	35.9	-26.1	37.1		
			-18.9	25.7	2.9	1.0	-21.0	26.4	-22.7	26.9	-24.3	27.5	-26.4	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.3	-1.4	2.2	8.2	13.5	18.9	21.6	20.7	16.9	10.8	5.0	-0.3
	DBStd	9.66	6.23	5.39	4.85	4.50	3.86	3.45	2.81	3.06	3.66	4.41	4.77	4.98
	HDD10.0	1599	414	320	246	85	9	0	0	0	1	43	163	320
	HDD18.3	3523	672	552	499	304	158	36	6	12	71	236	401	577
	CDD10.0	1403	1	1	5	32	118	266	359	333	207	68	12	2
	CDD18.3	284	0	0	1	1	8	52	106	86	27	3	0	0
	CDH23.3	1997	0	0	3	20	113	366	781	563	136	16	0	0
	CDH26.7	489	0	0	1	6	27	94	203	134	24	2	0	0
Wind	WSAvg	2.9	3.2	3.2	3.5	3.4	2.9	2.5	2.2	2.2	2.6	2.7	3.0	2.9
Precipitation	PrecAvg	1240	90	80	110	106	99	106	101	113	116	106	102	111
	PrecMax	1590	190	198	260	223	196	234	195	264	281	369	169	172
	PrecMin	904	46	35	20	44	23	29	37	47	34	11	40	37
	PrecStd	168	33	35	50	46	40	54	41	57	63	74	31	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	14.7	20.3	27.1	30.1	32.0	32.8	32.4	29.1	25.4	19.6	15.9
		MCWB	12.7	9.7	12.9	16.4	18.5	23.9	24.2	24.0	22.3	19.5	15.0	13.6
	2%	DB	12.2	11.0	15.9	22.2	27.1	29.1	31.0	29.7	27.1	22.4	17.0	12.2
		MCWB	10.9	7.9	9.4	13.4	18.0	21.6	23.5	23.4	20.4	18.3	14.0	10.1
	5%	DB	8.2	8.0	12.3	18.7	23.7	27.5	28.9	27.8	24.7	20.8	14.7	9.3
		MCWB	6.5	5.0	8.2	11.7	16.3	21.0	22.4	21.7	19.5	17.7	12.0	7.1
	10%	DB	5.5	6.2	9.4	16.1	21.2	25.5	27.5	27.0	22.8	18.2	12.7	7.3
		MCWB	3.3	3.3	6.0	9.8	14.9	19.4	21.4	21.0	18.5	15.2	9.7	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	12.4	14.2	17.7	20.8	24.5	25.7	25.4	23.7	21.4	16.8	13.7
		MCDB	13.7	13.4	19.4	24.2	26.5	29.8	30.6	29.8	26.7	23.6	18.2	14.7
	2%	WB	10.7	8.5	11.5	15.2	19.3	23.2	24.6	24.2	22.3	19.7	14.9	10.7
		MCDB	11.5	9.9	13.9	20.1	24.4	27.6	29.1	28.0	24.8	21.6	16.4	11.4
	5%	WB	7.0	5.9	8.6	13.1	17.6	22.1	23.6	23.2	21.1	17.9	12.7	7.7
		MCDB	8.0	7.5	11.4	16.6	22.3	26.0	27.3	26.4	23.1	19.5	14.3	8.9
	10%	WB	3.6	3.5	6.4	11.5	15.8	20.9	22.6	22.3	19.8	15.6	10.4	5.1
		MCDB	5.2	5.6	9.6	14.7	20.0	23.5	26.0	25.2	22.5	17.8	12.5	7.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.5	9.1	10.2	10.8	10.0	10.2	9.8	9.9	9.9	8.5	7.6
		MCDBR	9.8	10.7	13.0	15.1	14.6	12.2	11.5	10.9	11.4	12.1	10.2	10.1
		MCWBR	9.1	8.7	9.0	8.8	7.8	6.4	5.8	5.6	6.4	7.7	8.4	8.9
	5% WB	MCDBR	9.7	10.1	12.1	12.4	12.8	10.4	9.6	9.1	9.2	9.6	9.2	9.0
		MCWBR	9.6	9.7	9.3	8.5	8.0	6.2	5.5	5.2	6.3	7.7	9.1	9.1
Clear-Sky Solar Irradiance	taub	0.297	0.308	0.331	0.359	0.403	0.449	0.465	0.440	0.371	0.338	0.311	0.296	
	taud	2.454	2.439	2.441	2.398	2.298	2.203	2.169	2.229	2.398	2.509	2.543	2.534	
	Ebn at Noon	859	904	919	911	874	831	813	822	864	859	840	835	
	Edh at Noon	76	90	101	114	129	142	146	134	105	84	70	66	
All-Sky Solar Radiation	RadAvg	1.85	2.71	3.70	4.75	5.31	5.78	5.94	5.25	4.32	2.90	2.03	1.52	
	RadStd	0.14	0.24	0.28	0.51	0.52	0.55	0.33	0.32	0.37	0.29	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	+57

Nomenclature: See separate page